HK IMO Training 2020-2021
Problem Set 12 Suggested Solutions

Problem 45. Let a, b, c be positive real numbers such that $abc = 1$. Prove that

$$\frac{1}{a^4+b^3+c^2}+\frac{1}{b^4+c^3+a^2}+\frac{1}{c^4+a^3+b^2}\leqslant 1.$$

Solution. By the Cauchy-Schwarz inequality, we have

$$(a^4+b^3+c^2)(1+b+c^2)\geqslant (a^2+b^2+c^2)^2.$$

Similarly, we obtain

$$\begin{aligned}\frac{1}{a^4+b^3+c^2}&\leqslant\frac{1+b+c^2}{(a^2+b^2+c^2)^2},\\ \frac{1}{b^4+c^3+a^2}&\leqslant\frac{1+c+a^2}{(a^2+b^2+c^2)^2},\\ \frac{1}{c^4+a^3+b^2}&\leqslant\frac{1+a+b^2}{(a^2+b^2+c^2)^2}.\end{aligned}$$

Adding these, it suffices to prove

$$3+a+b+c+a^2+b^2+c^2\leqslant (a^2+b^2+c^2)^2.$$

By the AM-GM inequality and the QM-AM inequality, we have

$$3=3\sqrt[3]{abc}\leqslant a+b+c\leqslant\frac{1}{3}(a+b+c)^2\leqslant a^2+b^2+c^2.$$

Therefore,

$$3+a+b+c+a^2+b^2+c^2\leqslant 3(a^2+b^2+c^2)\leqslant (a^2+b^2+c^2)^2$$

as desired. □

Problem 46 (Bosnia Herzegovina TST 2017 Problem 3)**.** Find all real constants c for which there exists a strictly increasing sequence $\{a_n\}_{n\geqslant 1}$ of positive integers such that

$$\frac{a_{2n-1}+a_{2n}}{a_n}=c \qquad (1)$$

for any positive integer n.

Solution. The answer is any positive integer $\geqslant 4$.

It is clear that c is a positive rational number. Let $c=\dfrac{s}{t}$ with $(s,t)=1$. From (1), we know that $t\mid a_n$ for any n. Then $t\mid a_{2n-1}+a_{2n}$, and hence $t^2\mid a_n$ for any n from (1). Inductively, we find that $t^k\mid a_n$ for any positive integers k and n. This is only possible when $t=1$. So c is a positive integer.

When $c=3$, the relation becomes $a_{2n-1}+a_{2n}=3a_n$. Note that

$$\begin{aligned}a_1+a_2&=3a_1,\\ a_1+a_2+a_3+a_4&=3a_1+3a_2=3^2a_1,\\ a_1+a_2+\cdots+a_8&=3a_1+3a_2+3a_3+3a_4=3^3a_1,\\ &\ldots,\\ a_1+a_2+\cdots+a_{2^k}&=3^ka_1\end{aligned}$$

for any $k\geqslant 1$. Since the sequence is strictly increasing, this yields

$$3^ka_1=a_1+a_2+\cdots+a_{2^k}\geqslant a_1+(a_1+1)+(a_1+2)+\cdots+(a_1+2^k-1)=2^ka_1+2^{k-1}(2^k-1),$$

which implies $a_1 \geqslant \dfrac{2^{k-1}(2^k-1)}{3^k-2^k} \geqslant \dfrac{2^{2k-2}}{3^k} = \dfrac{1}{4}\left(\dfrac{4}{3}\right)^k$. This is a contradiction since the right-hand side goes to infinity when $k \to \infty$. Therefore, $c \geqslant 4$.

When $c \geqslant 4$, we define a sequence as follows: $a_1 = 1$, $a_2 = c-1$, and for any $n \geqslant 2$, let $a_{2n-1} = \dfrac{ca_n}{2} - 1$ and $a_{2n} = \dfrac{ca_n}{2} + 1$ if ca_n is even and $a_{2n-1} = \dfrac{ca_n - 1}{2}$ and $a_{2n} = \dfrac{ca_n+1}{2}$ if ca_n is odd. Clearly, $\dfrac{a_{2n-1}+a_{2n}}{a_n} = c$. It remains to prove $\{a_n\}$ is strictly increasing. We prove this by induction.

The base case $a_1 < a_2$ is obvious. The case $a_{2k-1} < a_{2k}$ holds as well. We now prove $a_{2k} < a_{2k+1}$. Note that

$$a_{2k} \leqslant \frac{ca_k}{2} + 1 \leqslant \frac{c(a_{k+1}-1)}{2} + 1 \leqslant \frac{ca_{k+1}}{2} - 1 \leqslant a_{2k+1}.$$

This shows $a_{2k} < a_{2k+1}$ unless all are equalities. The third equality holds when $c = 4$. But in that case we can prove inductively that the sequence is just $1, 3, 5, 7, \ldots$, so it is still strictly increasing. So we are done. □

Problem 47 (Greece TST 2015 Problem 4)**.** Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$yf(x) + f(y) \geqslant f(xy) \tag{2}$$

for any real numbers x and y.

Solution. Putting $x = 1$ in (2), we get $yf(1) + f(y) \geqslant f(y)$, i.e. $yf(1) \geqslant 0$. As this holds for both positive and negative y, we must have $f(1) = 0$. Putting $x = -1$ in (2), we get

$$yf(-1) + f(y) \geqslant f(-y). \tag{3}$$

Putting $y = -1$ and replacing x by y in (2), we get

$$-f(y) + f(-1) \geqslant f(-y). \tag{4}$$

Adding (3) and (4), we obtain $(y+1)f(-1) \geqslant 2f(-y)$. Replacing y by $-y$, this means

$$(1-y)f(-1) \geqslant 2f(y). \tag{5}$$

Next, putting $x = 0$ in (2), we get $yf(0) + f(y) \geqslant f(0)$. Together with (5), we obtain

$$(1-y)f(-1) \geqslant 2f(y) \geqslant 2(1-y)f(0).$$

As this holds for both positive and negative $1-y$, we must have $f(-1) = 2f(0)$. This shows all equalities hold. In particular, we have $f(y) = (1-y)f(0)$ for any $y \in \mathbb{R}$. One checks that any function $f(y) = c(1-y)$ where c is a constant is a solution to (2) since

$$yf(x) + f(y) = cy(1-x) + c(1-y) = c(1-xy) = f(xy).$$

□

Problem 48 (Bulgaria MO 2017 Problem 5)**.** Let $P(x)$ be a polynomial with real coefficients whose (complex) roots are distinct positive real numbers. Is it possible that the polynomial $Q(x) = x(x+1)(x+2)(x+4)P(x) + c$ is the kth power of a polynomial with real coefficients for some positive integer $k \geqslant 2$ and real number c?

Solution. No. It suffices to prove that $Q(x)$ is not a prime power of a polynomial with real coefficients. For odd $k \geqslant 3$, suppose $Q(x) = R(x)^k$ where $R(x) \in \mathbb{R}[x]$. Then $R(\alpha) = \sqrt[k]{c}$ for any root α of $x(x+1)(x+2)(x+4)P(x)$. There are $d+4$ such roots where $d = \deg P$. As $\deg R = \dfrac{1}{k}\deg Q = \dfrac{d+4}{k}$, we must have $\dfrac{d+4}{k} \geqslant d+4$. This is impossible.

It remains to show that $k = 2$ does not work. Suppose $Q(x) = R(x)^2$ where $R(x) \in \mathbb{R}[x]$. Then $c = Q(0) \geqslant 0$ and

$$x(x+1)(x+2)(x+4)P(x) = R(x)^2 - c = (R(x) + \sqrt{c})(R(x) - \sqrt{c}).$$

Let S_1 be the set of all roots of $R(x) + \sqrt{c}$, and let S_2 be the set of all roots of $R(x) - \sqrt{c}$. WLOG assume $-4 \in S_1$.

If $-2 \in S_1$, let C be the leading coefficient of $R(x)$, and let $a_1, a_2, \ldots, a_t$ be all the roots of $R(x) - \sqrt{c}$, which are all $\geqslant -1$. Then

$$C(-2-a_1)(-2-a_2)\cdots(-2-a_t) = R(-2) - \sqrt{c} = -2\sqrt{c} = R(-4) - \sqrt{c} = C(-4-a_1)(-4-a_2)\cdots(-4-a_t).$$

This is impossible since $|-2-a_j| = 2 + a_j < 4 + a_j = |-4-a_j|$ for each j. So we have $-2 \in S_2$.

If $-1 \in S_1$, using the same argument, we have

$$C(-1-a_1)(-1-a_2)\cdots(-1-a_t) = C(-4-a_1)(-4-a_2)\cdots(-4-a_t).$$

But then one of the a_j's is -2 and the others are $\geqslant 0$. So the two sides have different signs, contradiction. So we have $-1 \in S_2$.

If $0 \in S_1$, using the same argument, we have

$$C(0-a_1)(0-a_2)\cdots(0-a_t) = C(-4-a_1)(-4-a_2)\cdots(-4-a_t).$$

We may assume $a_1 = -1$ and $a_2 = -2$. Then $(-a_1)(-a_2) = 2 < 6 = (-4-a_1)(-4-a_2)$ and $|-a_j| = a_j < 4+a_j = |-4-a_j|$ for any other j. Again, the two sides cannot be the same. So we have $0 \in S_2$.

Now, for any two consecutive roots a_j and a_{j+1} of $R(x) - \sqrt{c}$, there exists a root of $R'(x)$ between a_j and a_{j+1} by Rolle's theorem. By counting the number of roots, we find that there is exactly one root of $R'(x)$ in this interval. In particular, there exists a unique root of $R'(x)$ between -2 and -1, and another root between -1 and 0. But the same holds for $R(x) + \sqrt{c}$. This means there exists a unique root of $R'(x)$ between -4 and the smallest positive root of $R(x) + \sqrt{c}$. This contradicts with the two roots found between -2 and 0. So $R(x)$ does not exist. $\square$